Quantum Cryptography for Securing the Internet of Things: A Comprehensive Literature Survey

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***Abstract*— The Internet of Things (IoT) now underpins critical systems like healthcare monitors, smart home, smart grids, automotive V2V & V2X networks, and industrial controls but it still relies on classical cryptography, whose security assumptions are vulnerable to quantum computers. A large quantum computer could run Shor's algorithm to break RSA/ECC and use Grover's algorithm to accelerate brute-forcing of symmetric keys, undermining the confidentiality and authentication of data in these IoT applications. Researchers have proposed quantum cryptographic solutions. Quantum key distribution (QKD) and quantum random number generators (QRNGs) exploit quantum mechanics to detect eavesdropping and deliver provably secure key material. For example, chip-scale QRNGs can be embedded in IoT sensors to generate true random keys, boosting resistance to brute-force attacks. These quantum techniques provide information-theoretic secrecy, for instance, QKD-generated keys have been used to authenticate smart-grid SCADA. However, most IoT devices are resource-constrained, making direct deployment of conventional QKD hardware challenging. Post-quantum cryptography (PQC) complements quantum methods by providing algorithms such as lattice-based, hash-based that resist quantum attacks on existing hardware. Standardized PQC can be deployed via software updates on legacy IoT nodes, enabling scalable and quantum-safe security within current constraints. Researchers are exploring lightweight PQC for connected cars and e-health sensors. Hybrid approaches combining QKD/QRNG with PQC offer a layered defense. Since QKD protocols are regarded as 'future-proof' and PQC standards mature, their combined use will be essential for robust, quantum-safe IoT security. Miniaturizing quantum components such as integrated QKD/QRNG modules and optimizing lightweight PQC schemes for low-power IoT devices are future research directions to secure IoT networks.**

Keywords—Quantum Cryptography, Post-Quantum Cryptography (PQC), Internet of Things (IoT) Security, IOT Security Threats, Quantum Key Distribution (QKD), Quantum Random Number Generator (QRNG)

I. INTRODUCTION

The IoT is growing at a very fast rate and is linking billions of devices in smart homes, industries, and cities. These devices are constantly capturing and transferring sensitive information thus the security of these devices is a top priority [3]. Classical cryptography algorithms like RSA, ECC, and AES are used in classical traditional IoT security for authentication and encryption. Nonetheless, these cryptosystems were threatened by quantum computing in a way that has never been seen before. A quantum computer with enough computing power would be able to factor or solve discrete logs in the time frame of a public-key algorithm (e.g. RSA/ECC) in the past [5][15]. Indeed, it was demonstrated in 1994 through a groundbreaking quantum algorithm by Shor that RSA and DiffieHellman would be compromised in case a large scale quantum computer is constructed [5]. There are estimates of the arrival of the Q-day (when quantum computers will crack the encryption of public keys) ranging between a few years and decades [13][15]. Even symmetric algorithms such as the AES are not immune. The algorithm introduced by Grover is capable of accelerating brute-force key search, effectively reducing the security level by half e.g. AES-128 becomes 64-bit strong [1]. Thus, the quantum-safe security has been a pressing issue in the IoT community.

Two general strategies have been developed to counter this looming threat. The first one is quantum cryptography which uses the laws of quantum physics to encrypt the communications in such a way that it cannot be decrypted even with an unlimited computing power. As an example, quantum key distribution (QKD) employes quantum states (photons) in order to generate encryption keys [17][18]. Quantum cryptography offers unconditional security regardless of computational assumptions [6][17]. The second is post-quantum cryptography (PQC), which is new classical algorithms thought to be resistant to quantum computer attacks [11]. PQC does not need quantum hardware, in contrast to quantum cryptography. It may be applied to traditional devices though it frequently has an increased cost in computation. Both methods are currently being sought after to make IoT security future-proof.

This survey paper presents a detailed overview of the state of the art in quantum cryptography of IoT security, compared to the classical and post-quantum methods. We use recent literature to present studies to highlight (i) IoT security weaknesses and limitations, (ii) quantum cryptographic schemes (e.g. QKD) that can be used with the IoT, (iii) post-quantum cryptographic algorithms (iv) existing industrial applications and testbeds, and (v) future challenges and directions. We also cover comparative analysis of the classical versus quantum cryptography in the IoT scenario. It aims to explain how quantum cryptography can supplement or substitute conventional security on IoTs to survive the quantum computing age [3][21].

II. BACKGROUND: QUANTUM COMPUTING & CRYPTOGRAPHY

A. Quantum Computing Threats to Classical Cryptography:

Security of classical cryptographic algorithms are based computational complexity e.g. the complexity of factoring large integers or discrete logarithms. Quantum computing does away with this paradigm. The algorithm of Shor can factor integers and solve discrete logs in polynomials on a quantum computer [5][15], which directly breaks RSA, DSA, Diffie Hellman and elliptic-curve cryptosystems. Symmetric-key cryptography (AES, 3DES, etc.) and hash functions, on

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the other hand, are relatively more resistant to quantum attacks, but they are not entirely immune. The algorithm presented by Grover offers a quadratic speed-up of brute force search, which means that doubling key sizes is needed to achieve the same security [1]. These threats have driven a rapid research on post-quantum algorithms and motivated institutions such as NIST and NSA to develop a migration plan to cryptography resistant to quantum attacks [15]. In July 2022, NIST announced the first set of PQC algorithms, one general encryption scheme and three digital signature schemes for standardization [13], a milestone toward transitioning away from vulnerable classical algorithms.

B. Post-Quantum Cryptography (PQC)

PQC consists of classical algorithms designed to resist attacks by quantum computers [11]. The PQC include lattice-based cryptography, code-based cryptography, hash-based signatures, and multivariate polynomial cryptography [11]. These rely on problems like lattice Shortest Vector Problem (SVP), decoding random linear codes, or multivariate quadratic equations that currently have no efficient quantum or classical solutions. For example, lattice-based schemes such as CRYSTALS-Kyber (a KEM/encryption scheme) and CRYSTALS-Dilithium (digital signature) emerged as finalists in the NIST PQC competition and are built on hard lattice problems [13]. Isogeny-based cryptography (supersingular isogeny problems) is another promising approach; it offers relatively small key sizes compared to other PQC branches [2], though some isogeny schemes (e.g. SIKE) were recently broken classically. PQC enables a mostly seamless upgrade of classical systems without new physical infrastructure, which is appealing for IoT devices. Several works survey the landscape of quantum-resistant cryptosystems for IoT [20], detailing candidate algorithms and performance on constrained devices [8].

C. Quantum Cryptography

In parallel, quantum cryptography takes a different route by using quantum-mechanical phenomena to secure data. The typical example is Quantum Key Distribution (QKD). QKD is a technique to securely share encryption keys using quantum states. In QKD, two parties e.g., Alice and Bob send quantum bits (qubits), typically polarized photons, over a quantum channel to establish a shared secret key. By quantum laws like the Heisenberg uncertainty principle and the no-cloning theorem, any eavesdropper e.g., Eve attempting to intercept the qubits will inevitably disturb their state, introducing detectable anomalies [17][18]. Thus, QKD provides *information-theoretic security*. As long as the quantum channel is not compromised, the keys generated are provably secure against even an adversary with unlimited computational power [17]. This is fundamentally different from classical key exchange e.g. Diffie–Hellman, which can be broken given sufficient computational resources. Bennett and Brassard's pioneering 1984 BB84 protocol established this concept [17], and since then many QKD protocols e.g. BB84 variants with 4 or 6 states, decoy-state QKD, E91 entanglement-based QKD, etc. have been developed to improve efficiency and robustness [14]. Besides QKD, quantum cryptography is also concerned with quantum state protocols such as quantum coin tossing, secret sharing and digital signatures [4][17].

Table 1 provides an overview of major dissimilarities between classical and quantum cryptography in the IoT. This comparison explains the reason why quantum methods are perceived to be an effective complement or replacement to classical methods in securing the future IoT systems.

TABLE I. COMPARISON OF CLASSICAL AND QUANTUM CRYPTOGRAPHY FOR IOT SECURITY

Aspect	Classical Cryptography (IoT)	Quantum Cryptography (IoT)
Security Basis	Computational hardness assumptions (e.g. factoring, discrete log) [18]. Security is conditional on attacker's computational limits.	Physics principles (quantum uncertainty, entanglement, no-cloning) [17][18]. Security is unconditional (independent of computing power).
Vulnerability to Quantum	Most public-key algorithms are breakable by Shor's algorithm [15]. Symmetric ciphers need increased key lengths due to Grover's algorithm.	QKD and quantum protocols are not known to be breakable by quantum computers (eavesdropping alters quantum states, revealing attacks) [18].
Key Distribution	Requires pre-shared secrets or public-key infrastructure which is quantum-vulnerable. Key exchange e.g. DH/ECC can be intercepted by quantum adversaries.	Performed via QKD. Keys are generated with one-time pad level security, and any eavesdropping attempt is detected in real time [17] [18].
Implementation	Software-based; runs on conventional processors. Lightweight crypto available for constrained IoT nodes, but future updates needed for PQC.	Requires quantum hardware (single-photon sources, detectors, possibly quantum random number generators). Integration with IoT devices is challenging due to size, cost, and power constraints.
Current Maturity	Mature and widely deployed (standard algorithms, libraries, protocols like TLS). Well-	Emerging technology; QKD is commercially available in niche markets (fiber/satellite links) but not yet

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	understood performance on IoT devices.	common in consumer IoT. Mostly in testbeds and pilot projects [14].
Scalability in IoT	Scales well in large networks using PKI for key management. Small footprint implementations exist for 8-bit/32-bit MCUs.	Scaling to many devices is non-trivial. Quantum channels needed for each link or a trusted node network. Quantum repeaters and networks (quantum internet) are in early research stages.

The classical cryptography is practical and can be implemented in the current IoTs but it will be insecure with the future quantum computers. The promise of quantum cryptography is a much greater security guarantee, but requires additional hardware and logistics.

III. IoT Security Threats and Limitations

Securing IoT devices is not an easy task in itself, and quantum threats make it even more difficult. The IoT devices are resource-constrained i.e. limited memory and CPU and battery power and they are prone to both cyber-attacks and physical attacks, as these devices are usually deployed either in unprotected/remote locations or by individuals who might physically tamper with them. The most prevalent attacks against IoT systems are impersonation of the device, listening to communications, man-in-the-middle attack, replay attack, and denial-of-service (DoS) disruption [24]. Compared to the conventional IT systems, IoT devices do not necessarily have frequent software updates or high-security authentication, which leads to numerous vulnerabilities that are reported in the sector [1]. To address some of the problems, lightweight cryptographic algorithms and protocols have been proposed, at the cost of strength versus efficiency, and may be easily defeated by a sophisticated attacker.

These threats are magnified by the imminent emergence of quantum computing. A survey by Lohachab *et al.* [1] notes that *"IoT security faces heightened risks with the advancement of quantum computing"* as quantum computing becomes a reality. Classical encryption schemes already strain constrained devices, and once large-scale quantum computers are realized, many of today's security protocols could quickly become ineffective.

A. Resource Constraints

IoT nodes often use 8-bit or 32-bit microcontrollers and have strict power budgets e.g. battery-powered sensors. Classical encryption like RSA/ECC already strains some constrained devices, leading to adoption of lighter primitives i.e. ECC over RSA, or even symmetric-key only schemes [8]. Post-quantum algorithms typically have much larger key sizes or higher computational complexity than classical ones, which could overwhelm these tiny devices. For example, experiments by Malina *et al.* [8] and Mansoor *et al.* [10] showed that lattice-based key exchange schemes like NewHope and Frodo take on the order of tens of seconds to execute on Raspberry Pi-class hardware which is a prohibitively long latency for many real-time IoT applications. Memory and network bandwidth can also be an issue. Many PQC schemes have kilobyte-size keys and ciphertexts, which is problematic for low-power wide-area IoT networks with small MTUs.

B. Key Management at Scale

IoT deployments involve large numbers of devices, often communicating peer-to-peer or with cloud services. Classical IoT security architectures typically rely on a server-centric model or pre-shared keys for local device-to-device links. Introducing quantum cryptography into this mix is non-trivial. QKD traditionally works for point-to-point links and would require either a quantum network infrastructure or a trusted node architecture to distribute keys network-wide [19]. Managing keys for millions of IoT endpoints is already challenging with PKI. Doing so with quantum keys which have short lifetimes and cannot be stored long-term without risk presents new research questions.

C. Lifecycle and Update Challenges

Many IoT devices have a long operational life i.e., 5–15 years and minimal maintenance. Devices being sold today may still be in use when large quantum computers come online. This exacerbates the risk, as vulnerable crypto baked into hardware or firmware today could be exploited in the future. The "Harvest now, decrypt later" attack [25] in which an adversary records encrypted traffic now and decrypts it later once a quantum computer becomes available. Upgrading cryptography on deployed IoT devices may be difficult due to infrequent or insecure update mechanisms. Thus, there is a window of exposure if we do not begin transitioning IoT to quantum-safe algorithms well ahead of Q-day [1].

IV. Quantum Cryptography Techniques for IoT

Quantum cryptography offers several techniques that could strengthen IoT security, the most prominent being Quantum Key Distribution (QKD). In principle, QKD allows two IoT devices or an IoT device and a gateway to establish secret keys with provable security. Once a secret key is generated via QKD, it can be used in standard symmetric encryption e.g. AES etc. to protect IoT data with unconditional security [16]. QKD does not replace encryption itself, rather, it replaces the vulnerable public-key exchange or key agreement process with a quantum-safe method [16][19]. This approach addresses one of the weakest points in IoT communication i.e., key exchange over insecure channels by making it highly resistant to eavesdropping attempts.

A. Applying QKD in IoT

Researchers have begun exploring how QKD can be integrated into IoT networks. One approach is to use a central quantum-enabled base station or satellite that distributes keys to IoT nodes. Dhar *et al.* [16] propose a novel architecture where a low-Earth-orbit satellite provides QKD keys to IoT devices, which then use a lightweight cipher like the ASCON AEAD algorithm for data encryption. The feasibility of satellite-assisted quantum key distribution has been experimentally demonstrated in real-world settings, where intercontinental quantum networks were established through satellite relays [22]. Figure 1 illustrates an IoT security architecture integrated with quantum components [16]. Qubits are distributed from a satellite to IoT gateways, enabling the generation of shared secret keys through QKD. These keys are

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then utilised with lightweight symmetric ciphers such as ASCON for efficient data encryption. To ensure end-to-end authenticity and integrity, the encrypted data is simultaneously recorded on a blockchain. This hybrid model demonstrates how quantum techniques can be practically combined with established mechanisms to provide confidentiality, authenticity, and tamper-evidence for IoT communications. This combination of quantum and classical techniques aims to leverage the strengths of quantum for key distribution and classical for efficient data encryption and integrity logging i.e., blockchain. Experimental or commercial implementations of such hybrid models are still in early stages, but they illustrate a possible path for deploying quantum security at scale.

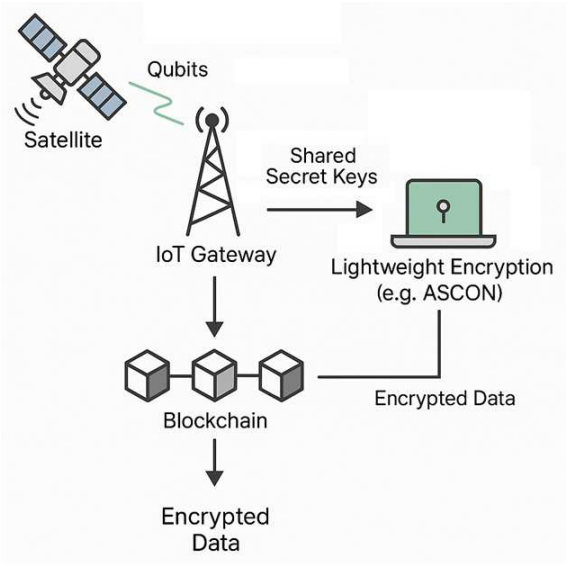


Fig. 1. IOT Security Architecture with Quantum Integration.

For local IoT networks e.g. within a smart home or a vehicle, fiber-optic QKD links or even free-space optical QKD could be used between high-value nodes. For example, an industrial IoT setup might use QKD between a control server and critical sensors/actuators in the field to prevent command injection or eavesdropping [1]. The laws of physics guarantee that any interception of the quantum signals by an attacker will be detected by legitimate nodes due to changes in measured qubit statistics, prompting the discard of those keys [7][18]. In this way, QKD can provide a level of trust that classical key exchange cannot.

However, directly applying QKD to IoT faces practical hurdles. QKD hardware i.e., single-photon emitters, detectors, etc. is currently expensive, power-consuming, and often requires precise conditions i.e., low noise, stable fiber links or line-of-sight for free-space. IoT devices, in contrast, are cheap, low-power and deployed in diverse environments. One possibility is to offload quantum operations to an edge device or gateway that has quantum capabilities, while the resource-limited IoT nodes interface with that gateway via classical links.

B. Quantum Random Number Generation (QRNG)

Many IoT devices rely on Pseudo-Random Number Generators (PRNGs) for cryptographic keys and nonces, which can be weak or predictable. A quantum random number generator (QRNG) generates random numbers based on quantum uncertainty e.g. measuring the noise of a quantum vacuum or radioactive decay to produce truly random bits. Bhatt *et al.* [14] propose using QRNG to create unique cryptographic keys for each IoT device, dramatically increasing the entropy and unpredictability of keys. Unlike general QKD, QRNG does not require two-party communication. It can be a standalone chip that feeds randomness into IoT cryptographic operations. Modern QRNG chips have been developed that are compact and could be integrated into IoT hardware, for example, Toshiba and ID Quantique offer USB-stick QRNGs. By equipping devices with quantum-generated keys, one ensures that even if an attacker has significant computational resources, they cannot guess or reproduce the keys. Bhatt *et al.* [14] note that the only way to obtain such a key would be to physically access the device, which is often impractical. QRNG-enhanced keys can significantly strengthen IoT security against both classical and quantum adversaries, as they introduce unpredictability that is practically impossible to replicate.

C. Quantum Authentication and Signatures

Beyond key distribution, researchers have explored quantum-based authentication and digital signature schemes relevant to IoT. Classical digital signatures like RSA, ECDSA etc. will be broken by quantum computers, and with post-quantum signature schemes, quantum physics offers alternative approaches too. One example is a quantum blind signature. Wen *et al*. [4] introduced a blind signature protocol using entangled EPR pairs and one-time pad encryption with quantum keys. In their scheme, the signer can sign a message without learning its content (blindness), and the security is guaranteed by quantum one-time pad encryption and QKD, rather than by number-theoretic assumptions [4]. They proved this quantum blind signature to be unconditionally secure and even argued it was feasible with then-current technology [4]. While this specific protocol might not be directly deployable in IoT yet, it demonstrates the potential for quantum-based authentication mechanisms that could replace vulnerable classical signatures in firmware updates, device onboarding, and transactions between IoT devices. Similarly, quantum key authentication methods verify device identities in ways that cannot be forged by adversaries without quantum capabilities. Chawla et al. [12] also emphasize quantum cryptography as a promising future in the protection of wireless sensor networks, which are a core part of the IoT ecosystems.

D. Device-Independent Quantum Security

A noteworthy concept for IoT is device-independent quantum cryptography. It is a protocol that remain secure even if the devices used have manufacturing flaws or backdoors. This is relevant because many IoT devices or their cryptographic modules are manufactured by third parties, and one must trust that they are implemented correctly. Device-independent QKD (DI-QKD) leverages quantum nonlocality to detect any deviation from expected behavior, and also it does not require trust in the internal workings of the devices. The IoT survey by Bhatt *et al.* [14] mentions that *"device-independent quantum cryptography can be used to ensure that the manufactured devices are trustworthy"*. This means an IoT device could potentially use a quantum protocol to prove its outputs are genuine and not influenced by a hidden Trojan. DI-QKD and related protocols are extremely demanding requiring nearly ideal conditions and detection efficiencies, but ongoing research may make them practical

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in the future, which would be a boon for supply-chain security in IoT.

To put these approaches into perspective, we compared the major cryptographic paradigms currently discussed for IoT environments. As shown in Table II, classical algorithms such as RSA, ECC, and AES are efficient in terms of latency, memory footprint, and energy consumption, but they become ineffective once quantum adversaries are considered. Post-quantum algorithms including Kyber and Dilithium provide resistance to quantum attacks, yet they impose higher computational overheads and larger key sizes, which directly affect energy usage and handshake latency on constrained IoT hardware [8][10][13]. In contrast, quantum cryptographic techniques such as QKD and QRNG offer unconditional security, though their reliance on specialised photon sources, detectors, and secure channels makes their integration into everyday IoT devices highly challenging [15].

TABLE II. COMPARATIVE ANALYSIS OF CRYPTOGRAPHIC APPROACHES FOR IOT ENVIRONMENTS

Metric	**Classical Cryptography (RSA/ECC, AES)**	**Post-Quantum Cryptography (Kyber, Dilithium, etc.)**	**Quantum Cryptography (QKD, QRNG)**
Energy Efficiency	Optimised libraries consume minimal energy on 8-bit and 32-bit controllers	Higher energy demand; lattice-based schemes often require 20–30% more power on microcontrollers [8]	Energy intensive due to photon sources and detectors
Latency	Millisecond-level handshakes and encryption	Medium latency (hundreds of ms); some lattice schemes show handshake times above 800 ms [10]	High latency; key distribute-on setup time significant
Memory Footprint	Small key sizes (32–128 bytes keys) fit into constrained RAM	Larger keys (1–2 KB); memory occupation may exceed 50% on small devices [8][10]	Minimal local memory; keys externally generated but requires buffer for secure exchange
Security Against Quantum Attacks	Vulnerable; RSA/ECC breakable by Shor's algorithm, AES reduced strength under Grover's	attacks, though side-channel risks remain [15]	physical principles of quantum mechanics
Deployment Readiness	Mature, widely used in IoT (TLS, DTLS, Zigbee)	Transition phase; NIST standardization (Kyber, Dilithium) completed in 2022 [13]	Early stage; available in pilot projects and niche markets (fiber/satellite QKD)
Scalability in IoT Networks	High scalability with PKI and lightweight symmetric schemes	Scalable with effort; communication overhead and large key sizes still challenging	Limited scalability; requires quantum channels, trusted nodes, or repeaters

This comparative assessment highlights that no single approach fully satisfies IoT's diverse security and performance needs. Instead, a hybrid design, where PQC algorithms protect general device-to-cloud communication and quantum methods secure high-value links, appears to be a practical compromise for near-term deployments.

V. POST-QUANTUM CRYPTOGRAPHIC ALGORITHMS FOR IOT

Post-Quantum Cryptography (PQC) refers to new cryptographic algorithms that are designed to be secure against quantum attacks. In 2016, NIST initiated an open competition-like process to evaluate and standardize PQC algorithms, drawing 82 submissions worldwide [13]. By 2022, this process produced a shortlist of winners for key establishment, a lattice-based KEM named CRYSTALS-Kyber was selected, and for digital signatures, CRYSTALS-Dilithium (lattice-based), FALCON (lattice-based), and SPHINCS+ (hash-based) were chosen [13]. These algorithms will be standardized and rolled out as replacements for RSA, Diffie–Hellman, ECDSA, etc.

Researchers have actively measured the performance of PQC on microcontrollers. Malina *et al.* [8] implemented several PQC key exchanges on 32-bit ARM Cortex-M devices and found that schemes like NewHope and Frodo which are lattice-based could complete key generation and shared-key computation in under a second on a 32-bit MCU, but with significant CPU usage.

Another major challenge with IoT devices is communication overhead. If an IoT device must send a 2 KB PQC public key during a handshake as opposed to a 32-byte ECC public key today, protocols need to accommodate that. For battery-powered devices on networks like Zigbee or NB-IoT, more transmitted bytes directly translate to more energy consumption. Thus, engineering efforts are needed to compress keys or use techniques like hybrid key exchange

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i.e., combining classical and PQC to reduce size until networks evolve.

Despite these challenges, PQC can be considered as a practical necessity for IoT in the near term. Some authors have already proposed PQC-based schemes tailored to IoT. Mansoor *et al.* [10] introduce an authentication scheme for IoT-based e-health systems using lattice-based cryptography within the TLS 1.3 protocol. Their scheme, termed "PQCAIE", relies on the hardness of the lattice SVP problem and integrates with the standard TLS handshake to provide quantum-resistant mutual authentication between IoT devices and servers. Early experiments suggest that the scheme is secure and computationally efficient on a standard IoT hardware [10]. This finding indicates that lattice-based PQC can be executed within the constraints of IoT systems and be integrated into the existing security frameworks.

Chawla et al. [3] and Liu et al. [20] researched on systematic comparison of the classical vs. post-quantum vs. quantum cryptography in IoT. They compare quantum attack key size, data size and time complexities of different schemes. Lohachab et al. [1] found that most classical protocols like RSA and ECC become ineffective during the quantum age. Post-quantum schemes can sustain security at the cost of larger key sizes and computational overheads that may not be feasible for tiny IoT nodes [1]. For example, AES-128 is relatively resistant to quantum attacks when used with 256-bit keys i.e. AES-256, but substituting ECC with authentication could mean using Dilithium or Falcon signatures many kilobytes in length, which is a larger jump [1].

It has been noted that a lot of classical schemes are ineffective in quantum era, including RSA and ECC. In comparison, post-quantum schemes are secure but tend to have trade-offs in the form of increased key sizes, increased latency and increased memory consumption.

A. Performance Metrics of PQC in IoT

The recent PQC algorithm benchmarking on constrained IoT platforms has shown that there are very essential trade-offs in terms of energy consumption, execution latency, and memory footprint. Malina et al. [8] have also found that lattice-based systems like Frodo and NewHope used almost 20-30% more energy than ECC on 32-bit ARM Cortex. On the same note, Mansoor et al. [10] reported that PQCAIE over an e-health IoT platform had an average latency of 850 ms/handshake, but was still feasible to medical telemetry. Another bottleneck is memory usage. in Kyber and Dilithium protocols, the size of public keys can be 1-2 KB which consumed up to 60% of available RAM on some microcontrollers [8][10]. These results highlight that the optimization of lightweight and hardware-based acceleration is necessary before PQC could be applicable across all IoT settings.

B. Side-Channel Considerations

The major concern of PQC susceptibility to side-channel attacks [9]. PQC algorithms often have complex arithmetic which might leak information via timing, power consumption, or electromagnetic emanations. Shaller *et al.* [15] specifically review side-channel attacks on the final NIST PQC candidates and countermeasures. They note that attackers can do cache timing attacks, fault injections, and power analysis on certain lattice-based implementations, recovering secret keys, making side-channel attacks easier. Thus, any PQC deployed on IoT must incorporate side-channel resistant techniques such as constant-time algorithms, masking, etc., but this will further increase the resource requirements.

Post-quantum cryptography provides a toolkit of quantum-safe algorithms that can be gradually integrated into IoT devices and protocols. While they preserve existing workflows, their increased computational and communication demands pose challenges for constrained IoT environments. Quantum cryptography can be used where it is necessary and if required, it can also be combined with PQC algorithms as a complementary layer. If one layer say, lattice-based encryption is broken in the future, another layer i.e., quantum-distributed key or hash-based signature can prevent a total security failure.

C. Industrial and Commercial Implementations

The advantages of quantum cryptography and the development of PQC algorithms is leading to real-world trials and products. This section highlights the state of industrial and commercial efforts in quantum-safe IoT security.

D. Quantum Key Distribution Networks

Several companies and research consortia have built quantum communication networks that may lay the groundwork for quantum-secured IoT. For example, Toshiba, ID Quantique, and QuantumCTek offer QKD devices that can generate and distribute keys over fibre optic links with applications in government and financial communications. In the telecom sector, SK Telecom (South Korea's telecom) has partnered with quantum technology companies i.e., ID Quantique, Telefónica, Toshiba to develop QKD for its 5G network infrastructure [1]. The goal is to use quantum-distributed keys to secure communication between 5G base stations and core network elements, which indirectly benefits IoT since 5G is a key enabler for massive IoT connectivity. These initiatives indicate that the telecom industry sees quantum cryptography as a viable upgrade to network security in the coming years.

E. Integrated Solutions Combining Blockchain and Quantum

In the IoT context, blockchain can be used for data integrity and decentralized trust. Dhar *et al.* [16] implemented a prototype where IoT sensor data is encrypted with a QKD-derived key, then hashed and recorded on a blockchain for integrity. This type of system guarantees confidentiality through QKD + symmetric encryption as well as integrity through blockchain. Figure 2 is the lifecycle of quantum enhanced IOT data flow. The first step in this process is to generate high-entropy keys at the device level in a process known as quantum random number generation. These keys are then inherently hard to predict or to break in a brute-force attack. These keys are then extended to lightweight encryption techniques that can be used in resource-constrained devices, and thus efficiency is maintained at the expense of security. Lastly, the encrypted data is stored on a blockchain registry, which guarantees that any unauthorised change or alteration is promptly noticed. This lifecycle view illustrates how quantum techniques and distributed trust mechanisms can be synchronised to create a holistic defence model for IoT systems.

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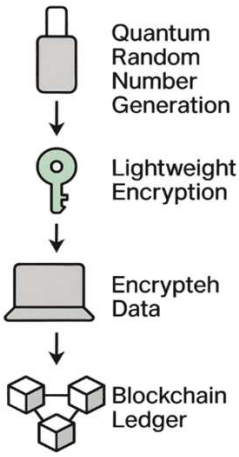


Fig. 2. Quantum-enhanced IOT data flow lifecycle

F. Firmware Signing and Update Security

One immediate application for post-quantum cryptography in IoT is securing firmware updates. Many IoT devices receive over-the-air updates that are signed by the vendor to ensure authenticity. If classical signatures like ECDSA are compromised by a quantum attacker, an adversary could potentially push malicious firmware to millions of devices. Companies are also aware of this risk. Bosch and IBM have experimented with XMSS which is a hash-based signature scheme standardized by RFC for signing IoT firmware [26][27], since XMSS is stateful but quantum-resistant and has reasonably small signatures around a few kilobytes.

G. Standardization and Compliance

Internet Engineering Task Force (IETF) and IEEE are actively developing standards for PQC and quantum key distribution. The IETF has issued RFCs for stateful hash-based signatures (RFC 8391 for XMSS). IEEE 802.15.4 is a standard for IoT wireless communication used by Zigbee and Thread. Moreover, the IEC and NIST are also working on quantum-safe practices in industrial and utility control systems. In 2020, NIST published a report on "Migration to Post-Quantum Cryptography" advising organizations to begin inventorying their cryptographic dependencies and preparing for PQC upgrades [23].

H. Demonstrations of Quantum-Secured IoT Links

A project in 2016 at Indian Institute of Science combined a quantum random number generator with IoT sensor nodes to encrypt sensor data in a testbed, highlighting immediate attainable benefits of quantum tech (QRNG) without full QKD [14]. In 2019, China's QKD network was used to secure smart grid communications in a pilot program. Electric power grids are essentially large-scale IoT with sensors and controllers, and securing their data telemetry is critical. Quantum keys were used to encrypt SCADA communications, providing an extra layer of security for grid operations. These pilots show that as QKD and quantum devices can directly interface with IoT sensors/PLCs in such critical infrastructure.

I. Automotive IoT and V2V

The automotive industry, dealing with vehicle-to- vehicle (V2V) and vehicle-to-everything (V2X) communications which connect vehicles, roadside units, and cloud services is essentially an IoT ecosystem on wheels has also started looking at quantum-safe security. Twardokus *et al*. [28] addresses the challenges of implementing post-quantum cryptography (PQC) in vehicle-to-vehicle (V2V) communications and propose a hybrid authentication protocol that combines classical cryptography with PQC using Dilithium signatures for authenticating messages between cars, highlighting minimal additional latency and efficient spectrum usage.

J. Public Key Infrastructure (PKI) Migration

IoT widely relies on PKI (digital certificates issued by CAs) for device authentication. Companies like DigiCert is using hybrid certificates that contain both a classical RSA/ECC public key and a PQC public key deployed on IoT devices so that they can be validated by both classical and PQC methods in parallel [1].

VI. Challenges, Open Issues, and Future Scope

Despite significant progress, many challenges still remain for implementing quantum cryptography and post-quantum methods to secure the IoT. We discuss some open issues and future research directions.

A. Integration and Interoperability

One practical challenge is integrating quantum cryptography with existing IoT networks and protocols. QKD provides raw keys, but these need to be integrated into protocols like TLS, DTLS, MQTT, etc. securely. Ensuring interoperability between quantum-secured links and conventional network infrastructure is non-trivial. For example, if only part of a communication path is quantum-secured, such as between a device and a gateway, the system is only as strong as the weakest link. Any further hops have to be insured by classical PQC or extended QKD links. This requires a unified framework where quantum-generated keys can be directly transferred to classical protocol stacks effectively. According to the survey by Alléaume et al. [19], the focus is put on the analysis of the implications of the combination of QKD with other cryptography primitives in practice. They examined the case of applying QKD to periodically refresh symmetric keys in network links, and the use of QKD in multi-user networks where a large number of IoT nodes might desire any-to-any secure communication. Their conclusion was that while QKD is powerful, it does not operate in isolation. Robust architectures are needed to manage keys, user access, and failover to classical methods if needed. Future research should focus on quantum-classical hybrid security architectures tailored for IoT, which determine how and when an IoT system uses QKD vs. PQC vs. traditional crypto, maximizing security while balancing cost and performance.

B. Cost and Miniaturization of Quantum Hardware

For quantum cryptography to pervade IoT, the hardware involved i.e., photon sources, detectors, quantum random number generators must become significantly cheaper, smaller, and more power-efficient.

C. Distance and Mobility

Photons traveling through fibre suffer loss, limiting fibre QKD to on the order of 100–200 km without trusted repeaters. Free-space QKD can cover global distances using satellites but requires line-of-sight for ground stations and is sensitive to weather. Developing quantum repeaters which

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use quantum memory and entanglement swapping to extend QKD distances is an active research area and part of the vision for a future "quantum internet" [3][6]. Until quantum repeaters mature, practical QKD networks will rely on intermediate trusted nodes to extend range, which introduces points of vulnerability and the nodes must be secure. For IoT, which often communicates over long distances, fully quantum-secured paths may require a combination of satellite links and fibre, and careful placement of trust. Mobility poses another challenge; for example, how can QKD be reliably performed with a moving IoT node? Some experiments have shown QKD with moving drones or satellites, but reliably handling dynamic connection and disconnection of nodes in a quantum network needs more work.

D. Reliability and Key Rates

IoT applications such as industrial control or health monitoring often require real-time reliability. Current QKD key generation rates may act as a bottleneck in high-throughput IoT applications. While many IoT devices transmit small data packets, applications such as video monitoring require much higher bandwidth. In such cases, if QKD cannot generate sufficient key material, the system may temporarily fall back on less secure alternatives. Techniques like *"continuous-variable QKD"* which uses standard telecom coherent detection and can achieve higher rates are being researched to address this. Additionally, quantum cryptography protocols beyond QKD like quantum-secure direct communication (QSDC) have been theorized, where the data itself is transmitted quantumly with inherent security. Recent research by Zhang et al. [29] has demonstrated QSDC across 100 km fibre with the transmission of text and even voice data at a high level of security. Such direct quantum communication may be used in the future by IoT devices to send ultra-critical messages without the additional encryption process.

E. Human and Organizational Factors

Trust and adoption are often a problem that is not given due attention. Quantum cryptography and PQC are complicated and relatively new to the practitioners. Uncertainty, expert deficiency or fear of risking to adopt untested technology may also result in industry resistance or slow adoption. The development of easy-to-implement and understandable guidelines will be essential. In the case of PQC, the standards of NIST will come in handy since vendors usually abide by NIST. In the case of QKD, the ETSI Quantum-Safe Security Working Group and ITU are working on standards, which have the potential of boosting commercial confidence.

VII. CONCLUSION

The survey shows clearly that although classical cryptography forms the basis of modern IoT security, it is not resistant to quantum attackers. Post-quantum cryptographical algorithms provide an effective transition mode with the current hardware, albeit with increased latency and memory, and energy usage. Quantum cryptography, including QKD and QRNG, on the other hand, provide unconditional security based on physical principles, but require special equipment.

The most possible approach in the next decade would be a hybrid solution, where PQC controls most of the IoT connections, and QKD protects mission-critical nodes. The transition is already in progress, as ongoing standardization by NIST and pilot builds in the automotive, healthcare and telecom industries point to the fact that the transition is already being made.

Future research should be focused for miniaturizing the hardware of quantum devices, improving side-channel resistance of PQC implementations, and getting a smooth fit into IoT protocol stacks. When undertaken, these developments can help achieve a genuinely quantum-safe ecosystem of IoT that can withstand quantum and classical attacks.

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